

Authenticated context Latest check-in + comfort choices + support goal
Published exercise metadata + feedback/events + recent impressions
Journal text and journal embeddings are excluded.

1 Eligibility filtering

Deduplicate; remove explicit exclusions, uncomfortable items, metadata contraindications, and high-activation breath-hold conflicts.

2 Suitability rules: R

Match six check-in dimensions to metadata; adjust for activation and intensity; clip the rule score to $[0, 1]$.

3 Semantic matching: C and H

ONNX MiniLM embeddings compare current context and onboarding goal with eligible exercise descriptions. Label lexical fallback explicitly.

4 Collaborative signals: B

Use positive-similarity neighbours only with at least two common interactions and at least two qualifying neighbours. Otherwise use cold start.

5 Final ranking, diversity, and explanation

Blend scores; subtract recency; prefer support alignment; diversify categories; conditionally rotate the final slot; return up to four items with reasons.

Persist and display Request + ordered items + impressions
Practice events and owned feedback influence subsequent requests.
Short or empty eligible results are preserved.